CHRISTOPHER FU

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EDUCATION

University of California, Los Angeles (UCLA) Graduation Date: Jun. 2026
M.S. Data Science in Biomedicine Cumulative GPA: 4.00/4.00
Relevant Coursework: Foundations of DS (Probability and Statistics), ML Applications in Biomedicine, DS for Medical Imaging, Research
Awards: University of California, Los Angeles Warren Alpert Computational Biology and Artificial Intelligence Scholar, Dean's Honor List

University of California, Los Angeles (UCLA) Graduation Date: Jun. 2024
B.S. Data Theory Cumulative GPA: 3.79/4.00

PROFESSIONAL EXPERIENCE

Warren Alpert Computational Biology and AI Network **Los Angeles, CA**
Warren Alpert Computational Biology and AI Network Fellow Sep. 2025 - Dec. 2025
- Built an end-to-end pipeline combining longitudinal post-mortem MRI data, SynthSeg, and autopsy reports to quantify brain structure volumes and deformation rates across patients, linking imaging-based biomarkers with ground truth labels for neurodegeneration analysis
- Developed scalable analysis workflows including 3D segmentation validation, patient-level heatmaps of anatomical volumes and deformation trends, and a pair plot of autopsy variables to assess feature relationships, producing a dataset of 1300 rows for our cohort
- Applied LOOCV logistic regression to evaluate performance of brain structures in Alzheimer's Disease classification, achieving 65% accuracy with high F1 on select features while underscoring sparse, interpretable biomarkers and limitations of small, imbalanced cohorts

Kaiser Permanente **Pasadena, CA**
Applied Data and Analytics Master's Intern Jun. 2025 - Sep. 2025
- Integrated ETL, EDA + feature engineering in Python on 1K physician survey responses to identify demographic + engagement drivers; conducted A/B testing for survey experimental design and synthesized findings for CME research consultants to use in future AI Summit
- Fine-tuned Hugging Face transformer models (Few-Shot learning) on 950K dermatology patient messages, collaborating with a dermatologist to build a message classifier for triage (Macro F1 = 0.891) ready for Fall 2025 deployment to streamline clinical workflows
- Adapted Kaiser Permanente's primary care message classification model across 5 medical specialties, identifying low-volume underperforming classes; provided recommendations to enhance specialty-level classification accuracy for predicting patient messages

UCLA Biodesign **Los Angeles, CA**
Biodesign AI Fellow Sep. 2024 - Jun. 2025
- Queried, transformed, and cleaned using SQL on OB/GYN patient flow healthcare data from electronic health records (EHR) of 80,956 rows spanning 2022–2024; discovered bed usage, delivery, & departmental efficiency trends by exploratory analysis (line plot, bar plot)
- Designed, implemented, and evaluated an LSTM model to forecast OB capacity time-series trends, training on 2022–2023 data; achieved a test RMSE of 17.69, providing actionable insights to optimize resource and staff allocation, as well as enhance operational planning
- Collaborated with a team of six to investigate clinical research using a school's comprehensive data repository; developed a project strategy, communicated findings, and proposed actionable next steps, emphasizing opportunities to improve efficiency and data-driven decisions

Children's Hospital of Orange County (CHOC) **Orange, CA**
Data Science Intern Jun. 2023 - Aug. 2023
- Analyzed demographic disparities among diagnosed anxiety patients by constructing multiple linear regression models using Python's Statsmodels and SciPy libraries, achieving 77% accuracy; conducted t-tests and other statistical analyses to identify actionable insights
- Led a team of four data science interns in delivering data-driven insights to physicians through visually engaging presentations; created bar plots, box plots, model summary tables, and geographical heat maps to highlight trends and promote predictive measures in mental health
- Engineered and proposed detailed action plans for deploying advanced unsupervised machine learning techniques to uncover latent factors contributing to anxiety disorders in pediatric patients, supporting data-driven interventions and improved outcomes

SKILLS

Programming: Python (NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, Jupyter, Tensorflow, Keras), SQL (Azure Data Studio), R (ggplot2)
Data Analysis & Machine Learning: Regression, Classification, Unsupervised (Clustering), Neural Networks (LSTM, CNN), NLP
BI & Visualization Tools: Tableau, PowerBI, MS Excel, Matplotlib, Seaborn, MS Powerpoint, MS Word, MS Office, Dashboards